

Kathmandu BernHardt College

Tribhuvan University
Institute of Science and Technology



Software Project Management
Lab Report

On

“Bloomdale – An Ecommerce Website”

Submitted To

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Abstract

The project entitled "Bloomdale" is a website, which is aimed to make shopping easier for the people through a website in a systematic way. "Bloomdale" is a store with plants, its accessories, and seeds or seedlings. The Internet nowadays has been part of many people's day-to-day life. There is no doubt to say that it has made our life easier. It is also an important means of business, industry, and education. This website is probably useful to make people shop from their home itself. It contains information about different types of plants like ornamental plants, its price, accessories, and all the shopping details and so on. Observing the working of the store system and after going through it, I have to know that there are many operations, which they have to do manually. It takes a lot of time and causes many errors while data entry. Due to this, sometimes we often face many problems and face disputes with the customers. Well, this application tends to help customers from different parts of the world to view, search and buy some plants, gardening equipment, fertilizers and plant seedlings. Green Plants are the source behind natural cooling and for fresh environment, different ornamental plants play a vital role. It aims to take the already existing physical plant store to digital market space allowing more flow of customers and increase in revenue. It also focuses on making the shopping experience for customers easier, faster and convenient.

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Chapter 1: Introduction

1.1 Introduction

Ecommerce is fast gaining ground as an accepted and used business paradigm. More and more business houses are implementing websites providing functionality for performing commercial transactions over the web. It is reasonable to say that the process of shopping on the web is commonplace [1]. The main objective of this project is to develop a general-purpose e-commerce store where any product (such as books, clothes, mobiles, home appliances, coffee, plants and its machineries with some tools or accessories) can be bought from the comfort of home through the internet.

Online shopping is the process consumers go through to purchase products or services over the internet. Our site is one of those. This site provides items related to categories like gardening accessories, various indoor plants, outdoor plants, ornamental plants, pet friendly plants etc. It is an ecommerce project dedicated to people who are looking for products online and offline. Plants provide shade and beauty around home, schools, markets, places of work and parks. They also help to conserve energy and quality of air [2]. So “Bloomdale” is basically a store with different types of plants, its seedlings, equipment, and fertilizers. There are a number of options in today's world when people want to buy indoor or outdoor plants from nurseries/ plant shops. Either they can go to the physical nursery garden and clarify their queries regarding plants, or they can go to online shopping sites. So, it increases huge traffic in the store. Which would be beneficial for the store. The system helps to track the prices of different products across various e-commerce platforms [3].

1.2 Objectives

The main target of building this site is to provide the customer wider options for the desired items with reasonable prices. Furthermore, the general objectives can be summarized as:

- It saves the time and labor of the customer.
- To build a system with an application in real world scenarios.

Chapter 2: Process Model

The waterfall model for a plant online store is a structured approach to software development that progresses through sequential stages, each building upon the completion of the previous one. In the context of a plant online store, this model begins with the initial stage of requirements gathering and analysis, where the specific needs and objectives of the online store are identified. This involves understanding the types of plants to be sold, features required for online shopping such as browsing, search functionality, checkout process, and integration with payment gateways.

Following requirements analysis, the design phase entails creating a blueprint for the online store's architecture, user interface, and database structure. Design considerations include user experience, visual aesthetics, navigation flow, and backend systems for managing inventory, orders, and customer data. Once the design is finalized, the development phase commences, where the actual implementation of the online store takes place. Developers build and integrate the frontend and backend components according to the specifications outlined in the design phase.

After development, the system undergoes rigorous testing to ensure functionality, usability, and security. This testing phase involves various types of testing such as unit testing, integration testing, system testing, and user acceptance testing. Any issues or bugs discovered during testing are addressed and resolved before proceeding to the next phase. Once testing is complete and the online store is deemed ready for deployment, it is launched to the public.

The final phase of the waterfall model involves maintenance and support, where ongoing updates, enhancements, and bug fixes are performed to ensure the continued smooth operation of the online store. This phase also includes monitoring user feedback and analytics to identify areas for improvement and optimize the online shopping experience. The waterfall model provides a systematic approach to developing a plant online store, emphasizing thorough planning, sequential progression, and attention to detail throughout the software development lifecycle.

Chapter 3: Cost-Benefit Analysis

A cost-benefit analysis (CBA) is a process that is used to estimate the costs and benefits of decisions in order to find the most cost-effective alternative. A CBA is a versatile method that is often used for the business, project and public policy decisions. An effective CBA evaluates the following costs and benefits:

Costs are:

1. Direct costs
2. Indirect costs
3. Intangible costs
4. Opportunity costs
5. Costs of potential risks

Benefits are:

1. Direct
2. Indirect
3. Total benefits
4. Net benefits

The main purposes in using CBA:

- a) To determine if the project business case is sound, justifiable and feasible by figuring out if its benefits outweigh costs.
- b) To offer a baseline for comparing projects by determining which project's benefits are greater than its costs.
- c) Comparing two or more projects
- d) Deciding whether to start a new business
- e) Making the right hiring decisions
- f) Comparing investment opportunities
- g) Evaluating social benefits
- h) Evaluating policies
- i) Making decisions regarding changes

Cost-benefit analysis techniques are a common business activity owners and managers use to assess various projects. These techniques essentially compare the total capital investment

for the project against its potential returns. Several techniques are available, with the most common being the payback period, net present value, and rate of return. Companies can use one or all of the cost-benefit analysis techniques. The assessment occurs after the company has all necessary information and prior to investing capital into one or more of the projects. Net present value (NPV) is used to calculate the current value of a future stream of payments from a company, project, or investment.

- To calculate NPV, you need to estimate the timing and amount of future cash flows and pick a discount rate equal to the minimum acceptable rate of return
- The discount rate may reflect your cost of capital or the returns available on alternative investments of comparable risk.
- If the NPV of a project or investment is positive, it means its rate of return will be above the discount rate.

NPV analysis is used to help determine how much an investment, project, or any series of cash flows is worth. The net present value of a project is a measurement of profit that is calculated by subtracting the present values of cash outflows from the present values of cash inflows over a period of time. It is an all-encompassing metric, as it takes into account all revenues, expenses, and capital costs associated with an investment in its Free Cash Flow (FCF). The discount rate is used to calculate the present value of the future cash flows coming in or out of your project. In addition to factoring all revenues and costs, it also takes into account the timing of each cash flow that can result in a large impact on the present value of an investment. For example, it's better to see cash inflows sooner and cash outflows later, compared to the opposite. The mathematical formula for the calculation of the NPV is:

$$NPV = \frac{R_t}{(1 + i)^t}$$

where;

NPV = net present value

R_t = net cash flow at time

t i = discount rate t = time

of the cash flow

The advantages of NPV are:

1. NPV provides an unambiguous measure.
2. NPV accounts for investment size.
3. NPV is straightforward to calculate (especially with a spreadsheet).

In our cost benefit analysis, our project group is deciding to undertake a project which lasts for 5 years. The expected total project cost is 100000 and discount rate is 10%.

The calculation of NPV in excel is as follows:

SUM		X ✓ fx		=(NPV(B9,B3:B7)+B2)				
	A	B	C	D	E	F	G	H
1	Period(years)	Cashflow	Discount(10%)	Cumulative Discount				
2	0	-200000	-200000	-200000				
3	1	100000	90909.09091	-109090.9091				
4	2	50000	41322.31405	-67768.59504				
5	3	50000	37565.74005	-30202.855				
6	4	100000	68301.34554	38098.49054				
7	5	50000	31046.06615	69144.55669				
8								
9	Discount	10%						
10	Formula	=(NPV(B9,B3:B7)+B2)						
11	NPV	69144.56						
12								
13	Net Profit	150000						
14	Average Annual Profit	30000						
15	ROI (in %)	15						
16								
17	Payback period	3.44						
18	Total benefit with 10% dis.	269144.5						
19	Benefit-Cost Ratio	1.3457225						
20								
21								
22								
23								
24								

Figure 1: Cost Benefit Analysis

So, if discount rate is 10%, the project is worth the investment. Hence, our project group can undertake that project because $NPV > 0$. Along with NPV, net. Profit, annual average profit, return on investment, payback period and benefit cost ratio were also calculated and their values are NPR 150,000, NPR 30,000, 15%, 3.44 years and 1.35 respectively.

Chapter 4: Project Planning and Scheduling

Project planning entails selecting and designing effective policies and methodologies to achieve project goals. Project scheduling is the process of assigning tasks and allocating appropriate resources to complete them within an estimated budget and time frame. Project planning and scheduling are closely related and essential components of project management. Project planning and scheduling are both important aspects of project management that contribute to project success. Project managers can ensure that their team stays on track and completes the project on time and within budget by defining project objectives, developing a plan, and scheduling tasks. The stages of project planning are:

1. Select project
2. Identify project scope and objectives
3. Identify product infrastructure
4. Analyze project characteristics
5. Estimate effort for activity
6. Identify activity risks
7. Allocate resources
8. Review/publicize the plan
9. Execute plan
10. Lower-level planning

4.1 Activity Planning

A project plan is a schedule for activities indicating the start and stops for each activity. Project planning and scheduling require careful consideration of activity planning. It involves breaking the project down into smaller tasks or activities, figuring out how much time and resources will be needed for each activity, and deciding what order the activities should be completed in. Each activity should have some deliverables for ease of monitoring.

4.1.1 Work Breakdown Structure

A work breakdown structure (WBS) is essential for the development of a counterfeit money detection system as it helps break down the project into manageable tasks, milestones, and deliverables. By creating a WBS, the project team can clearly define the scope of work, allocate resources efficiently, identify dependencies between tasks, and establish a timeline for completion. This structured approach ensures that all aspects of the system development

are carefully planned and executed, leading to a more organized and successful project implementation.

4.1.1 Work Breakdown Structure

A work breakdown structure (WBS) is essential for the development of system as it helps break down the project into manageable tasks, milestones, and deliverables. By creating a WBS, the project team can clearly define the scope of work, allocate resources efficiently, identify dependencies between tasks, and establish a timeline for completion. This structured approach ensures that all aspects of the system development are carefully planned and executed, leading to a more organized and successful project implementation.

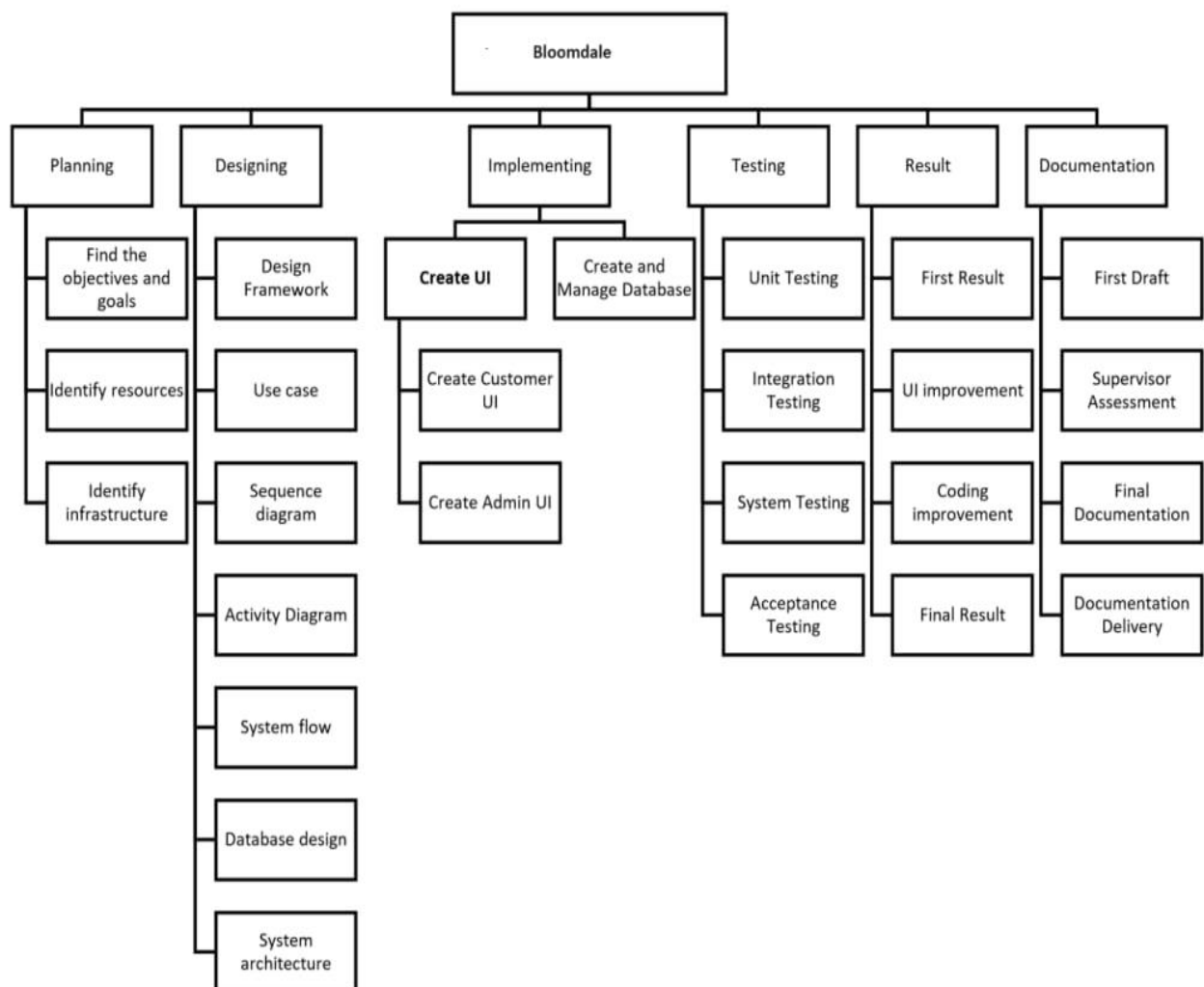


Figure 2: Work Breakdown Structure

	Task Name	Duration	Predecessors	Start	Finish
1	▸ Bloomdale	131 days		Mon 9/25/23	Mon 3/25/24
2	▸ Planning	10 days		Mon 9/25/23	Fri 10/6/23
3	Find the objectives and goals	6 days		Mon 9/25/23	Sat 9/30/23
4	Identify resources	3 days	3	Mon 10/2/23	Wed 10/4/23
5	Identify infrastructure	3 days	3	Wed 10/4/23	Fri 10/6/23
6	▸ Designing	54 days		Sat 10/7/23	Wed 12/20/23
7	Design Framework	1 day	5	Mon 11/13/23	Mon 11/13/23
8	Use case	1 day	7	Wed 12/13/23	Wed 12/13/23
9	Sequence diagram	1 day	7	Thu 12/14/23	Thu 12/14/23
10	Activity Diagram	1 day	7	Fri 12/15/23	Fri 12/15/23
11	System flow	1 day	7	Sat 12/16/23	Sat 12/16/23
12	Database design	1 day	11	Mon 12/18/23	Mon 12/18/23
13	System architecture	2 days	12	Tue 12/19/23	Wed 12/20/23
14	▸ Implementing	69 days		Wed 12/20/23	Mon 3/25/24
15	▸ Create UI	18 days		Wed 12/20/23	Fri 1/12/24
16	Create Customer UI	6 days	13	Thu 12/21/23	Thu 12/28/23
17	Create Admin UI	6 days	13	Thu 12/28/23	Thu 1/4/24
18	Create and Manage Database	6 days	16,17	Fri 1/5/24	Fri 1/12/24
19	▸ Testing	26 days		Fri 1/12/24	Fri 2/16/24
20	Unit Testing	4 days	18	Thu 2/1/24	Tue 2/6/24
21	Integration Testing	3 days	20	Wed 2/7/24	Fri 2/9/24
22	System Testing	4 days	20,21	Mon 2/12/24	Thu 2/15/24
23	Acceptance Testing	1 day	22	Fri 2/16/24	Fri 2/16/24
24	▸ Results	23 days		Fri 2/16/24	Tue 3/19/24
25	First Results	5 days	23	Mon 2/19/24	Fri 2/23/24
26	UI Improvement	6 days	25,23	Mon 2/26/24	Mon 3/4/24
27	Coding Improvement	8 days	25,26,23	Tue 3/5/24	Thu 3/14/24
28	Final Result	3 days	27	Fri 3/15/24	Tue 3/19/24
29	▸ Project Documentation	7 days		Sat 3/16/24	Mon 3/25/24
30	First Draft	2 days	28	Wed 3/20/24	Thu 3/21/24

Figure 3: WBS of Bloomdale in MS.Project

4.1.2 Critical path Method (CPM)

The Critical Path Method (CPM) is crucial for the development of a Bloomdale as it helps identify the sequence of tasks that determine the project's overall timeline. By using CPM, project managers can pinpoint the most critical tasks that directly impact the project's completion date. This allows for effective resource allocation, timely task prioritization, and proactive management of potential delays. By utilizing CPM, the project team can ensure that system is developed efficiently and on schedule.

Table 1: Critical Path Method

Activities	Abbreviation	Duration (Days)	Predecessors
Prepare Document or Memorandum	A	15	-----
Develop Project Charter	B	13	A
Blueprint & Budget Preparation	C	16	A
Resource Gathering	D	11	A
Develop Project Plan	E	15	B
Submit Project Plan	F	5	C
Procure Hardware/Software	G	10	D
Design Websites	H	14	D
Project Management	I	18	E, F
Testing	J	27	I, G
Audit Procedure	K	10	H
Update files & records	L	15	J, K
Closing the project	M	8	L

To form a critical path, $EF-LF=0$ AND $ES-LS=0$

Early Start(ES)	Activity	Early Finish (EF)
Late Start (LS)	Duration	Late Finish (LF)

Critical Path Method (CPM) Task Sequence:

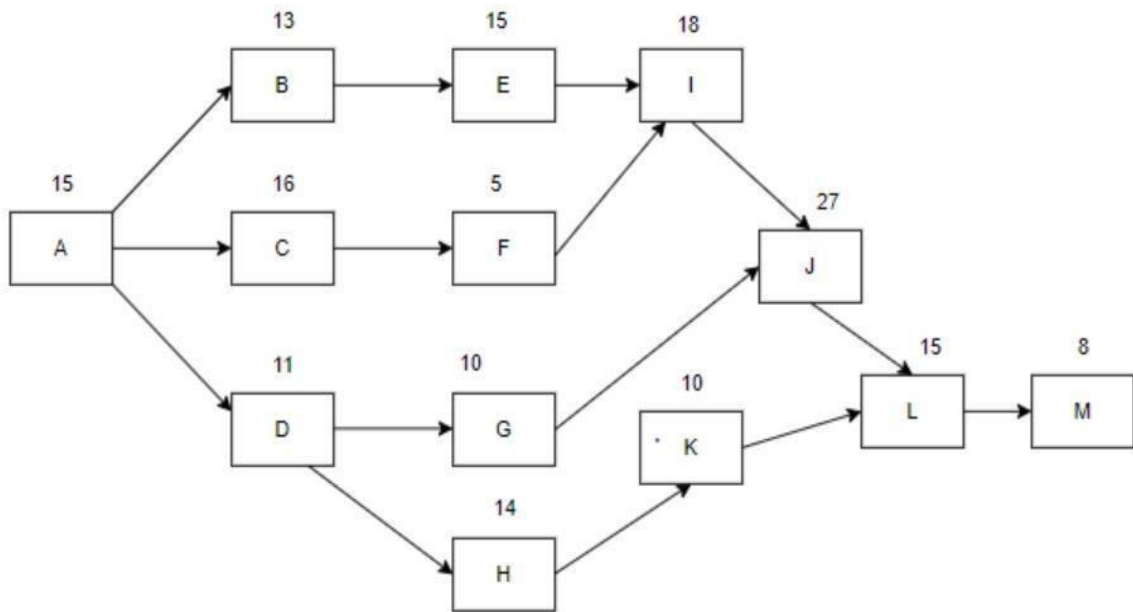


Figure 4: Critical Path Method

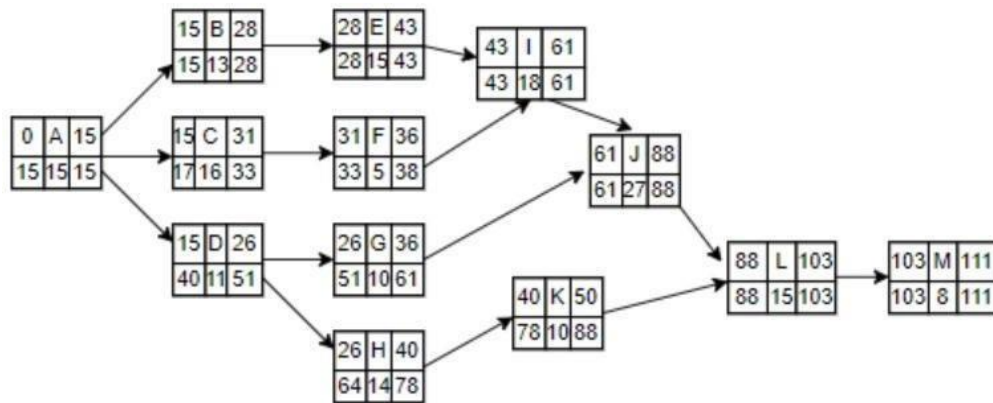


Figure 5: Node- Diagram of Sequence of Activities

Hence, “*A-B-E-I-J-L-M*” would form critical path. Critical Path Duration

$$= A+B+E+I+J+L+M$$

$$= 15+13+15+18+27+15+8$$

$$= 111 \text{ days}$$

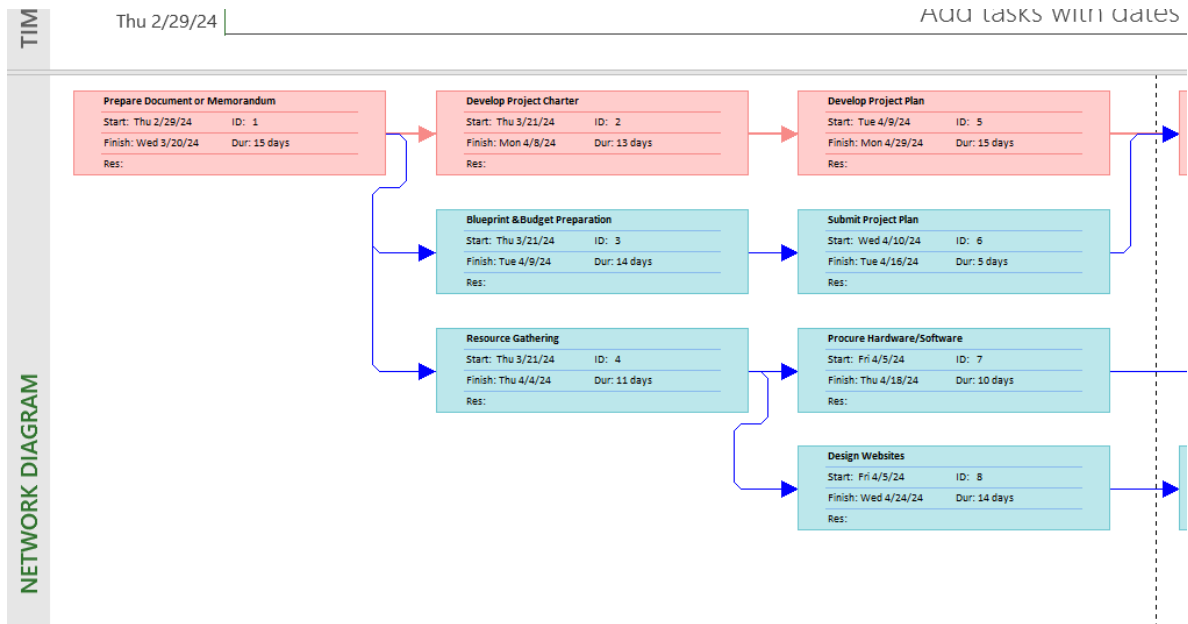


Figure 6: Network Diagram-1

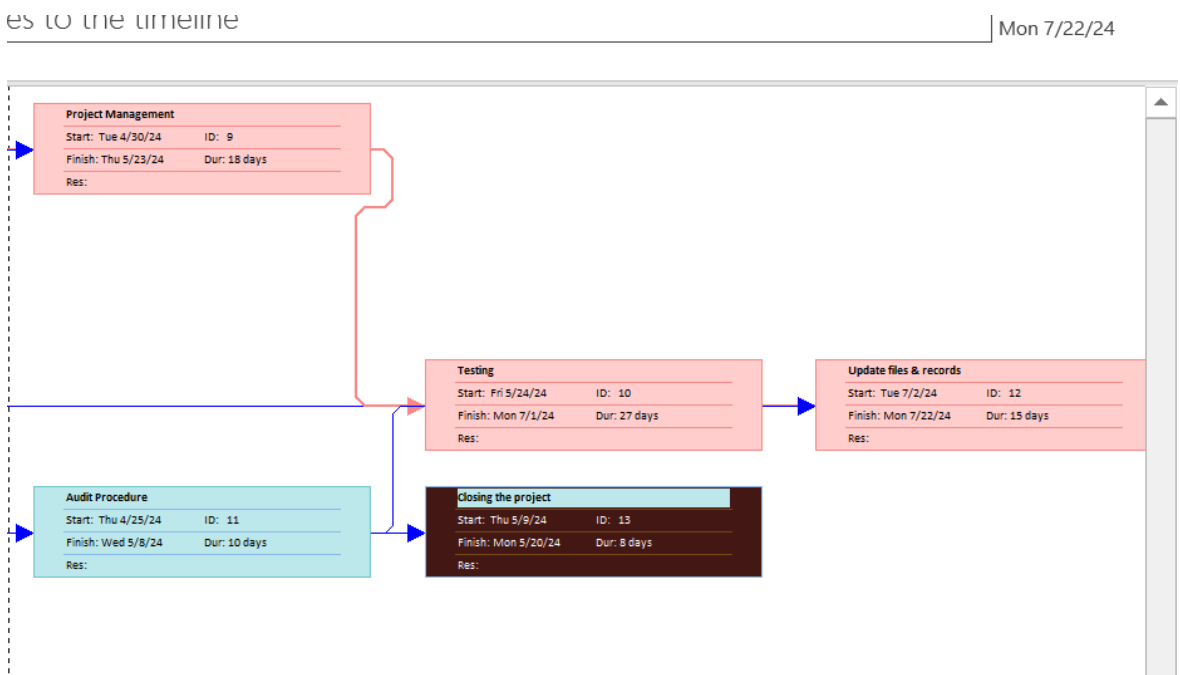


Figure 7: Network Diagram-2

4.1.3 Program Evaluation Review Technique (PERT)

PERT (Program Evaluation and Review Technique) is a project management tool used to analyze and manage the tasks involved in completing a project. PERT uses a network diagram to represent the tasks and their dependencies, and estimates the time required to complete each task. The network diagram is created by identifying the tasks required to complete the project and the order in which they must be completed. Once the network

diagram is created, PERT uses three-time estimates for each task: the optimistic time, the most likely time, and the pessimistic time. The optimistic time is the best-case scenario for completing the task, the pessimistic time is the worst-case scenario, and the most likely time is the most realistic estimate. Using these time estimates, PERT calculates the expected time for each task using the formula:

$$\text{Expected Time} = (\text{Optimistic Time} + 4 * \text{Most Likely Time} + \text{Pessimistic Time}) / 6$$

4.2 Project Schedule

Task scheduling is a crucial aspect of project planning, serving as a means to communicate and understand the necessary activities, allocated resources, and timeframes for completion. It plays a vital role in achieving project success, cost efficiency, and client satisfaction. Project scheduling in project management involves detailing tasks, deliverables, and milestones essential for project completion. Unlike standard weekly planner notes, project scheduling is more comprehensive, incorporating detailed information. The Gantt chart stands out as the most widely used and significant type of project timeline in project management.

4.2.1 GANTT Chart

A Gantt chart is a project management tool that aids in the planning and scheduling of projects of all sizes; they are especially useful for project visualization. A Gantt chart is a graphical representation of activity against time that aids project managers in monitoring progress.

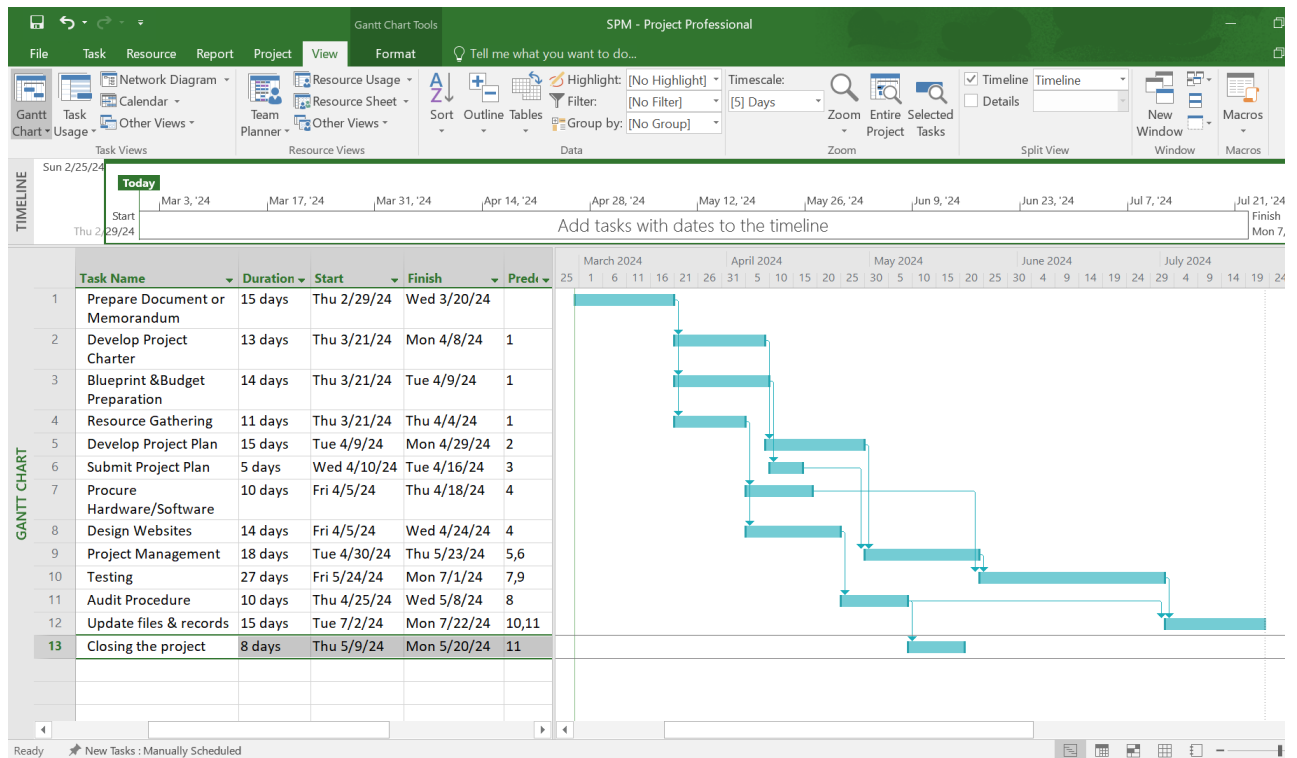


Figure 8: Gantt Chart View

4.2.2 Ball Chart

The way of showing whether or not targets have been met or not. It is represented in the form of circles that indicate the start and the end completion of activities. Circles of the ball chart mostly contain only two dates, the original and the revised one. An activity is denoted by a red circle and green color denotes that the activity is ahead of its schedule. Slippage in the project completion date but it is overcome by the timeline charts.

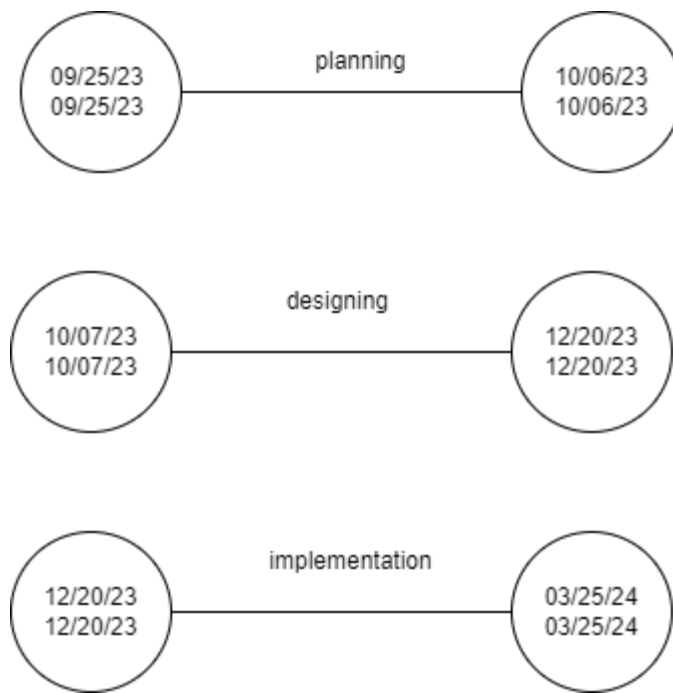


Figure 9: Ball Chart diagram

Chapter 5: Risk Analysis

5.1 Risks

Creating an ecommerce system has some risks that we need to be careful about. One risk is that using multiple authorization problem. Another risk is security threats. To avoid these risks, it's important to test the methods well, and use strong security practices.

Table 2: Risks, Impacts and Mitigating Strategy

Possible Risks		Impacts	Mitigating Strategy
1.	Data PrivacyConcerns	Potential legal and ethical consequences if not managed appropriately.	✓ Implement robust encryption, and conduct regular security audits.
2.	Technological Issues	Website downtime and performance issues can result in lost sales and damage the user experience.	✓ Invest in reliable hosting services and scalable infrastructure.
3.	Logistics and Fulfillment	Shipping delays and lost or damaged goods can result in customer dissatisfaction.	✓ Diversify logistics partners to minimize risk. ✓ Provide real-time tracking information to customers. Have a clear and customer-friendly return policy.
4.	Limited Time and resource constraints	Unavailability of the proper resources can lead to financial losses for businesses and individuals.	✓ Implementing and maintaining the algorithms may require significant time, expertise, and financial resources.

Chapter 6: Resource Allocation

6.1 Staffs and Equipment Required for Project

The Staff or Human Resource require in order to coordinate and complete the projects are: Project Manager, Software Developer, Testers, UI/UX Designers, and HR manager etc.

The Equipment or tools required for the project are:

- Electricity
- Hardware & Software Requirements

6.1.1 Staffs

Table 3: Staffs

Profession Title	Stakeholder's Name
Project manager	Mr. Abhimanu Yadav, Ashma Ghimire
HR Manager	Kabina G.C.
Programmers	Salina Gurung, Jeena Maharjan
Tester	Jeena Maharjan
UI/UX Designer	Ashma Ghimire, Salina Gurung

6.1.2 Hardware Tools

Table 4: Hardware Tools

Device	Laptop
RAM	4GB-8GB
Hard Disk	Min 32GB- Max 64GB

6.1.3 Software Tools

Table 5: Software tools

Name of Components	Specification
Operating System	Windows 10 and 11
Processor	Min 32GB- Max 64GB
IDE	VS Code
Database	SQ Lite
Backup & Repository	GitHub
Diagram Tools	Ms project
Processor	64bit, Intel Core 4, 2.5 GHz minimum or Pentium 630MHz

6.2 Total Cost and Duration of the Project

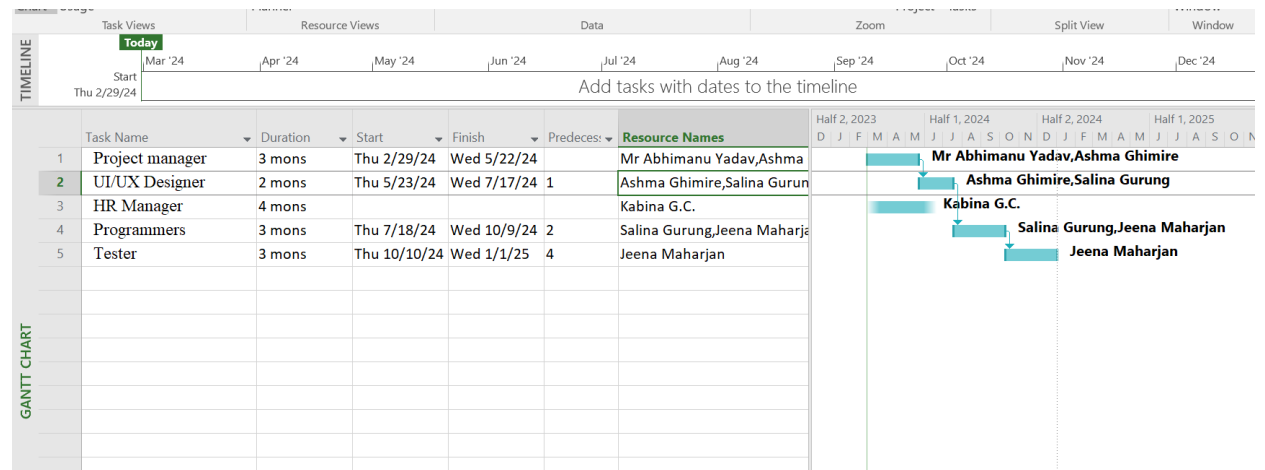
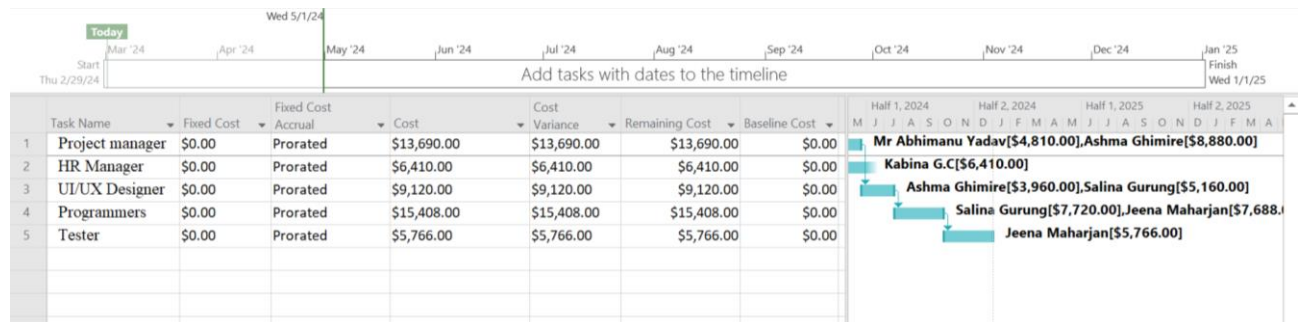
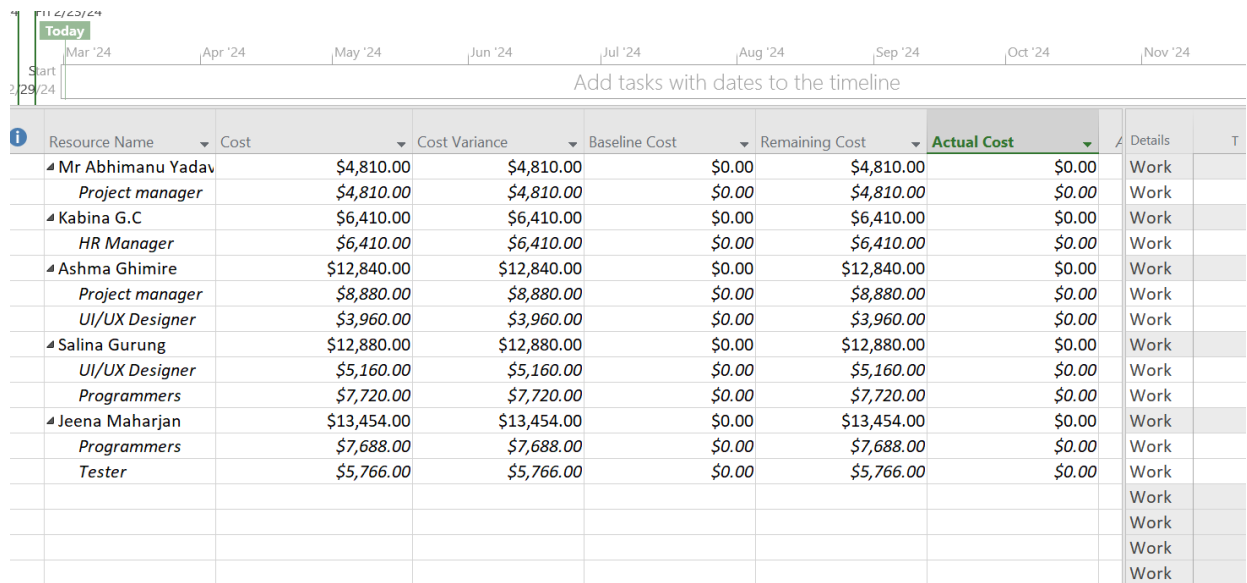


Figure 10: Human Resource Allocation



Hardware and Software				
S.No.	Name	Quantity	Price	Total Price
1	Laptop	5	75000	375000
2	Router	2	4500	9000
3	Printer	1	15000	15000
4	Multiplug	4	500	2000
5	Mouse	5	1000	5000
Total				406000

Figure 13: Cost estimation of Hardware and Software

Miscellaneous			
S.No.	Particulars	Monthly Expenses	Total Expenses
1	Bonus & Comission	5000	30000
2	Advertisement	15000	90000
3	Office Expenses	6000	36000
4	Repair & Maintenance	10000	60000
5	Travel & Entertainment	8000	48000
6	Utilities	10000	60000
Total			324000

Figure 14: Miscellaneous expenses

Consumable				
S.No.	Items	Duration	Price/Month	Total
1	Internet	6	1500	9000
2	Electricity	6	3000	18000
3	Generator	6	7000	42000
4	Office Rent	6	30000	180000
5	Transportation & Food	6	15000	90000
Total				339000

Figure 15: Consumable item cost estimation

Total Budget			
S.No.	Particulars	Duration of the Project	Total Cost
1	Staff	6	284000
2	Hardware & Software	6	406000
3	Consumable	6	339000
4	Miscellaneous Expenses	6	324000
Sub Total			1353000
VAT 13%			175890
Grand Total			1528890

Figure 16: Total Budget

6.3 Calculate if the project is on schedule

Solution, we have

Total Estimated or Baseline Budget = Rs. 1528890

Total Working Time Periods (Days) = 111 days [4 months (Oct-Feb)]

Budget Spend on Project = 55%

Completed/Work Planned Budget = 38%

So,

Let us assume that 45% of the work has already been completed. Earned value (EV) =

38% of Total Estimated or Baseline Budget

$= 45/100 \times \text{Rs. } 1528890$

$= 688000.5$

Planned value (PV) = 45% of Total Estimated or Baseline Budget

$= 38/100 \times \text{Rs. } 1528890$

$= 580978.2$

Actual cost (AC) = 55% of Total Estimated or Baseline Budget

$= 55/100 \times \text{Rs. } 1528890$

$= 840889.5$

Cost Performance Index (CPI)

$= \text{Earned value} / \text{Actual Cost}$

$= 688000.5 / 840889.5$

$= 0.81$

Schedule performance Index (SPI)

$= \text{Earned value} / \text{planned value}$

$= 688000.5 / 580978.2$

$= 1.18$

Based on 55% budget spent, The Schedule Performance Index (SPI) has greater value than the project that completed on time.

Cost Variance (CV) = Earned value – Actual cost

$= 688000.5 - 840889.5$

$= -152889$

The outcome result of Cost Variance (CV) is positive which means that the project is not running over the budget.

Schedule Variance (SV) = Earned value – planned value

= 688000.5 - 580978.2

= 107022.3

Hence, the value of Schedule.

It shows that value is schedule variance is 107022.3 then which is positive value the project is ahead.

Chapter 7: Implementation and Testing

7.1: Implementation

Systems development is the process of defining, designing, testing, and implementing a new software application. It can include the internal development of customized systems, the creation of database systems or the acquisition of third party developed software. There are 3 significant systems development techniques that been used to solve system problems.

The system advancement strategies are SDLC (Systems Development Life Process), JAD (Joint Application Development), and RAD (Quick Application Development). The tool, hardware and software used in making system are as follow:

- Tools
 - Database: MySQL
 - Backend: JavaScript, PHP
 - Frontend: HTML, CSS, Bootstrap
- Hardware requirement
 - Computer /Laptop
 - WIFI
- Software requirement
 - Office package
 - Visual Studio
 - XAMPP
- Development Method
 - Development method used to make this system is Waterfall Model of system development life cycle, where each phase has to be completed and signed off before subsequent phases can begin. This model is named “Water Fall” because its diagrammatic representation resembles a cascade of waterfalls. This model is easy to understand and reinforces the notion of “define” and “design” before “code”.

In the waterfall model, the project is split up into phases: requirements capture, analysis, design, implementation, testing, maintenance, etc., which means development moves from concept, through analysis, design, implementation, testing, installation, troubleshooting, and

ends up at operation and maintenance. Each phase of development proceeds in strict order, without any overlapping or iterative steps. In the strict case, each phase must be finished before development proceeds to the next phase.

7.1.1 HTML

HTML (Hypertext Markup Language) is the code that is used to structure a web page and its content. For example, content could be structured within a set of paragraphs, a list of bulleted points, or using images and data tables. HTML is a markup language that defines the structure of your content. HTML consists of a series of elements, which you use to enclose, or wrap, different parts of the content to make it appear a certain way or act a certain way. The enclosing tags can make a word or image hyperlink to somewhere else, can italicize words, can make the font bigger or smaller, and so on.

7.1.2 PHP

The PHP Hypertext Preprocessor (PHP) is a programming language that allows web developers to create dynamic content that interacts with databases. PHP is basically used for developing web-based software applications. PHP is a server-side scripting language that is embedded in HTML. It is used to manage dynamic content, databases, session tracking, even build entire e-commerce sites.

7.1.3 MYSQL

MySQL is an open-source relational database management system (RDBMS) with a client server model. RDBMS is a software or service used to create and manage databases based on a relational model. MySQL allows you to handle, store, modify and delete data and store data in an organized way. MySQL comes with an in-built tool known as MySQL Workbench that facilitates creating, designing, and building databases. MySQL offers support for multiple storage engines along with plug-in storage, making it more flexible.

7.1.4 XAMPP Server

XAMPP is the most popular PHP development environment. XAMPP is a completely free, easy to install Apache distribution containing MariaDB, PHP, and Perl. The XAMPP open-source package has been set up to be incredibly easy to install and to use. It is free and easy to use and easily available for Windows, Linux and Mac OS. It is a beginners friendly solution

package for full stack web development. It is an open-source software package which gives an easy installation experience.

7.2: System Testing

Software testing is a process of verifying and validating whether a software or application is bug-free, meets the technical requirements as guided by its design and development, and meets the user requirements effectively and efficiently by handling all the exceptional and boundary cases. It aims not only at finding faults in existing software but also at finding measure to improve the software in terms of efficiency, accuracy and usability. It helps to verify the functionality and performance of software application. Before actually implementing the new system into operation, a test run of the system has been done for removing the bugs. After codifying the whole program of the system, a test plan should be developed and run-on a given set of test data. The output of the test run should match the expected results.

Test ID: 1.1

Module Name: Login Page

Test Title: Verify the login with valid username and password.

Description: User will enter username and password. If it matches the data in the database then proceed. Else login is denied.

Table 6: System Test Case 1.1

Steps	Test step	Test data	Expected result	Actual result	Result (P/F)
1	Navigate to login	-	Fetching time less than 10s Inserting time less than 5s Page load time less than 3s	As expected,	P

2	Enter Username Enter Password	Username: salina Password: Salina12	Login should be successful	Successful login	P
3	Click login	-	System should generate homepage with user dashboard	As expected,	P

Test ID: 1.2

Table 7: System Test Case 1.2

Steps	Test step	Test data	Expected result	Actual result	Result (P/F)
1	Navigate to login	-	Fetching time less than 10s Inserting time less than 5s Page load time less than 3s	As expected,	P
2	Enter Username Enter Password	User: ashma Password: Ashma12	Login should be unsuccessful	Unsuccessful login,	P
3	Click login	-	Display message of incorrect password	Error message	P

Test ID: 2.1

Module Name: User Registration

Test Title: Verify if details are stored properly

Description: When a new user registers, all his/her details are entered in a database.

Table 8: System Test case 2.1

Steps	Test step	Test data	Expected result	Actual result	Result (P/F)
1	Navigate user to the Registration Page	-	Fetching time less than 10s Inserting time less than 5s Page load time less than 3s	As expected,	P
2	Enter Username Enter email Enter Password	User: salina12 Email: salina@gmail.com Password: salina123	Data should be added in the database	All the data is added to database.	P
3	Click Register	-	Registration should be successful	Registration Success	P

Test ID: 3.1

Module Name: Payment Options

Test Title: Verify payment method

Description: Whenever product has been bought, it has to be stored in the database and payment option will come and we have to choose payment method.

Table 9: System Test 3.1

Steps	Test step	Test data	Expected result	Actual result	Result (P/F)
1	Navigate to payment Page	-	Fetching time less than 10s Inserting time less than 5s Page load time less than 3s	As expected,	P
2	Method is Cash on delivery and order status is pending	-	Khalti button shouldn't be shown	Khalti button appears	F
3	Method is Esewa and status is completed	Esewa is selected	Esewa button should appear	Button appears	P

Chapter 8: Conclusion and Future Recommendation

8.1 Conclusion

Bloomdale is a company online system, which enables customers to check availability of plants, its accessories and fertilizers. It makes it easy for the customer to get plants and other materials which are related to growing plants. The online plant store is new in the context of Nepal but is surely making its impact felt. More people are switching to an online payment system as it is portable and more secure. We conclude that a web application will be developed which provides a better choice of products to customers and they can buy whatever they want. It becomes easier for us in our daily lives. Having a store on a digital platform also increases the revenue for physical stores and can become popular through the power of the internet. It will be a win-win situation for both the customers and business owner.

8.2 Future Recommendation

Ecommerce has become a basic property or element for any new upcoming and uprising businesses. Every people have their own personal devices where they just want the need of their desire with just the flick of their finger. So, with this website I want to deliver a good customer service and uplift their productive time. Effective modules can also be added from time to time.

In future, new categories might be added to the system whereas of now the customer only have two categories: Indoor and Succulent. More new features like linking of bank or UI upgrades or smoothness can be improved. We also can add new API integration for the payment portal and much more.

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